Ch. 2. The Natural Numbers as an Ordered Set.

Def. 2. Associated to a strict linear order $<$ on a set S, we define a nonstrict order relation $\leq$ by defining for all $x, y \in S$

$$x \leq y \overset{\text{def}}{\iff} x < y \text{ or } x = y.$$

mixed Transitivity

Thm. 2 If $x < y$ & $y \leq z$, then $x < z$.

Also, if $x \leq y$ and $y < z$, then $x < z$.

<u>Proof.</u> Suppose $x < y$ & $y \leq z$. We have two cases. $y = z$, $y < z$

If $y < z$, then transitivity shows $x < z$. ✓

Conversely, if $y = z$, then $x < y = z$, so $x < z$. ✓

§ 2.2 The Standard Order Relations on $\mathbb{N}$.

Def. 5. Given $m, n \in \mathbb{N}$ we write $m < n$ if there exists $b \in \mathbb{N}^+$ such that $n = m + b$.

Ex. $1+1=2$ from chapter 1. Thus $1<2$ since 1 plus a positive natural gives 2.

Theorem 4. If $x,y \in \mathbb{N}$ and if $y \neq 0$, then $x < x+y$.

Theorem 5. If $m \in \mathbb{N}$, then $m < m+1$.

Theorem 6. (Transitivity) Suppose $x,y,z \in \mathbb{N}$ and $x<y$, $y<z$. Then $x<z$.

Proof. Suppose $x,y,z \in \mathbb{N}$ and $x<y$, $y<z$. By def. 5, $y = x+b$ and $z = y+c$ for some $b,c \in \mathbb{N}^+$ (I.e. $b,c \neq 0$)

Thus $z = y+c = (x+b)+c \overset{\text{ch. 1}}{=} x+(b+c)$.

Now by ch. 1 $b+c \neq 0$, so $b+c \in \mathbb{N}^+$.

Thus $x<z$. $\square$

Lemma 7. If $n \in \mathbb{N}$, then $n=0$ or $0<n$.

Lemma 8 (Weak Trichotomy) Exactly one of the following holds $n=0$ or $0<n$.

If $m,n \in \mathbb{N}$, then at least one of the following hold.

$m<n$, $m=n$, $n<m$.

1. Transitivity Suppose $x, y, z \in \mathbb{N}$ and $x<y$, $y<z$. Then $x<z$.

2. (Trichotomy) exactly one of $x<y$, $x=y$, $y<x$ hold.

Thm. 9. $<$ is a strict linear order on $\mathbb{N}$.

Proof. We have shown $<$ is transitive (Thm. 6). By Lemma 8, we have weak trichotomy. If we can show $x \not< x$ for any $x \in \mathbb{N}$ (anti-reflexivity), then we are done Exercise 2! BWOC suppose $x<x$.

This means $x = x+b$ for some $b \in \mathbb{N}^+$. Now by ch. $x = x+0$, so we have $x+0 = x+b$. By Cancellation (ch. 1) $0=b$. Thus $0 \in \mathbb{N}^+$! However $0 \notin \mathbb{N}^+$. With this contradiction, we see $x \not< x$.

We conclude $<$ is a strict linear order. $\square$

Def. We say $x \leq y$ if either $x<y$ or $x=y$.

exclusive

Thm. 10 $0 \leq n$ $\forall n \in \mathbb{N}$ (for all)

Thm. 11. If $m, n \in \mathbb{N}$, then

$m \leq n \iff$ there exist $a \in \mathbb{N}$ with $m+a=n$.

$\exists$

Proof sketch. If $m<n$, then $a \in \mathbb{N}^+$ exists by def. of $<$.
If $m=n$, then $m = m+0 = n$, so $a=0$ works.

§2.3 Basic Properties of the order on $\mathbb{N}$

Def. If $m,n \in \mathbb{N}$, then we write $m>n$ if $n<m$.

Thm. 12. $n \neq 0 \Leftrightarrow n>0$.

Cor. 13. $n \in \mathbb{N}^+ \Leftrightarrow n>0$.

Thm. 14. Let $n \in \mathbb{N}$. There is no x such that $n<x<n+1$.

Proof. BWOC Suppose $n<x$ and $x<n+1$.

Thus $x = n+b$ for some $b \in \mathbb{N}^+$ and $x+c = n+1$ for $c \in \mathbb{N}^+$

Therefore

$$x = n+b = n+\sigma(d) = n+(d+1) = n+1+d = x+c+d.$$

since $b \in \mathbb{N}^+$, $b = \sigma(d)$ for some $d \in \mathbb{N}$.

ch. 1

ch. 1

Here $c+d \in \mathbb{N}^+$, so

$x < x$. This contradicts Trichotomy.

$\boxed{x = x!}$

We conclude there are no natural numbers between n and $n+1$. yay!

§2.4. The Cancellation Law for Mult.

Thm. 14. If $x, y, z \in \mathbb{N}$ such that $xz = yz$ and $z \neq 0$, then $x = y$.

§2.5. The Set $\{1, \ldots, n\}$

Def. Let $n \in \mathbb{N}$, then $\{1, \ldots, n\} \overset{\text{def}}{=} \{x \in \mathbb{N} : 1 \leq x \leq n\}$

Example $\{1, \ldots, 5\} = \{1, 2, 3, 4, 5\}$

$\{1, \ldots, 0\} = \{x \in \mathbb{N} : 1 \leq x \leq 0\} = \emptyset$

More General

Def. Let $m, n \in \mathbb{N}$, then $\{m, \ldots, n\} \overset{\text{def}}{=} \{x \in \mathbb{N} : m \leq x \leq n\}$

$\{3, \ldots, 2\} = \emptyset$

Notational Def. If $n \in \mathbb{N}$, then $\langle n \rangle = \{1, \ldots, n\}$.

⋮ ⋮ ⋮

Theorem 29. Every upper-bounded, nonempty subset of the naturals has a max.

Proof Sketch. Let T be such a set, and write B for an upper bound. Thus if $x \in T$, then $x \leq B$. Likewise $x \geq 0$ since $T \subsetneq \mathbb{N}$. We see

$$T \subset \{0, \dots, B\} = \{n \in \mathbb{N} : 0 \leq n \leq B\}.$$

Thm. 28 says T has a maximum.

Def. An ordered set U is <u>well-ordered</u> if every nonempty subset has a minimum.

Theorem 30. The natural numbers are well-ordered.

"The well-ordering Principle"

Remark. The well-ordering principle is equivalent to induction Axiom 4.

Proof. Let $S \subseteq \mathbb{N}$ be a non-empty subset. Let L be the set of lower bounds of S.

Note $0 \in L$ since $0 \leq x \ \forall x \in \mathbb{N}$ and $S \subseteq \mathbb{N}$.
In particular, $L \neq \emptyset$. Since $S \neq \emptyset$ there exists $k \in S$.
If $\ell \in L$, then $\ell \leq k$ since ℓ is a lower bound.
We see k is an upperbound for L. Thm 29 says L has a max; call it M. Greatest lower bound.
We wish to show $M \in S$. For a $\Rightarrow\Leftarrow$
Suppose $M \notin S$. Thus $M < k$ for every $k \in S$.
By Thm. 14 $M+1 \leq k$ for every $k \in S$.
$\rightarrow M < k < M+1$
Thus $M+1 \in L$ and $M < M+1$ contradicting maximality of M. $\times$ $M \in S$. $\square$